Collapsible Heels Using a Kresling Pattern

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References

Introduction

Heels are worn worldwide to increase height, enhance posture, and express style.
Typical users own multiple heel heights, leading to:

- High material consumption
- High cost
- Storage inefficiency

Problem: each heel height requires a separate shoe → wasteful, bulky, and unsustainable.
Our solution: a collapsible Kresling heel offering four height options in one shoe, eliminating the need for multiple pairs.
Key advantages:

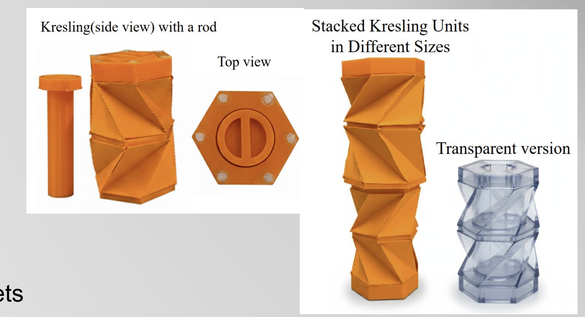
- Sustainable: reduces material use from four heels to one
- Cost-effective: hollow Kresling structure minimizes material
- Transformable: height adjusts through origami-based twist–extension motion



Design + Development

Natural Kresling advantages

- Axial extension/contraction
- Shell-based stiffness
- Compact collapsed height
- (Near) bistability for stable states



Overall Structure

- Hexagonal Kresling shell (m = 6)
- Central removable rod with magnets
- Top platform + bottom plate
- Two units of opposite chirality → cancels twist for user comfort

Geometric Formulation

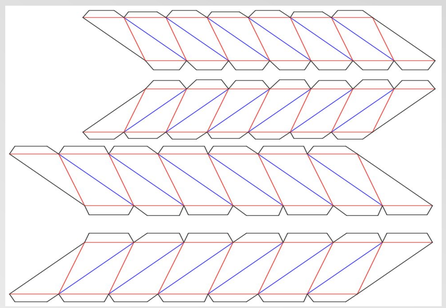
Parameters: a (top edge), b (bottom), β (crease angle), m = 6, Φ (twist), h (height), r (radius), chirality ∈ {±1}
MATLAB used to compute φ, h, vertex geometry*

<u>Large prototype:</u>

- a = 3 cm, b = 4.249 cm
- β ≈ 110.5°
- Twist φ ≈ –55° (≈120° offset)
- Height h ≈ 3.23 cm

<u>Small prototype:</u>

- a = 2.5 cm, b = 3.54 cm
- Same ratios preserved



Kresling Pattern Designed Using AutoCAD

Panels intentionally kept partially folded → increases lateral stiffness

Materials: 3D-printed panels, rod, magnets, paper walls
Tested: crease accuracy, magnetic rod stability, collapse smoothness, load stiffness
Iterations improved: magnet strength, chirality alignment, panel stiffness

Result

Structural Performance

- Stable vertical load-bearing with rod inserted
- Folded facets distribute force; opposite chiralities reduce sway
- Height matches MATLAB predictions
- Partially folded panels reduce local buckling risk



Transformability

- Smooth deploy ↔ collapse motion
- Collapse activated by stepping when rod removed
- Rod self-aligned via magnets
- Stackable modules → adjustable heel height



Limitations

- Sensitive to fabrication errors
- Stiffness depends heavily on central rod
- Collapsed height still tall due to joints & rod housing
- Full shoe integration (comfort, inclination change) still unsolved
- Needs improved lateral stiffness + more intuitive locking mechanism

Conclusion

- Not only can you wear paper, it can support your bodyweight!
- Blend style, comfort, and sustainability with the Kresling origami pattern
- Switch between bistable states to wear heels and flats with the same pair of shoes
- Fashionistas! Take the hidden support rod out of the center of the origami and step to twist the heel into its collapsed state–flats
- To return to heels, insert rod and twist for magnetic locking mechanism
- Voila! Save time, money, storage, materials and your feet!



Lesson Learned

- Kresling geometry highly sensitive to crease angle + measurement errors
- Small misalignments → major assembly issues
- Material stiffness trade-off:
- Thin → easy twist, but fragile
- Thick → durable, but harder to collapse
- Future work: test flexible polymers or composites for optimal performance